

JEET SARKAR

Software Engineer and Computer Science undergraduate specializing in full-stack development and AI-enabled backend systems.

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PROFESSIONAL SUMMARY

Software Engineer and Computer Science undergraduate specializing in **full-stack development** and **AI-enabled backend systems**. Experienced in designing and deploying **scalable web applications** with JavaScript/TypeScript, Python, React, Node.js, and Git. Developed **AI-driven search and recommendation engines** achieving sub-200ms latency and created autonomous coding and analytics platforms. Proven problem-solver with a global ranking in coding competitions, passionate about **developer tooling** and open-source ecosystems.

WORK EXPERIENCE

AI and Backend Development Intern | Internship

Jun 2026 - Present

Valantech

Hybrid • Kolkata

- Architected a **3-tier chatbot pipeline** (Redis cache lookup -> intent-matching layer -> LLM fallback) for web and mobile, **reducing average response latency by ~85%** (from 3s to sub-100ms) for cache-hit queries by serving pre-computed responses via Redis instead of invoking the LLM.
- Designed and led the **end-to-end architecture** of SIA, an in-house conversational AI for expense tracking and in-app support, integrating MCP servers to enable **autonomous execution of user-selected actions** — eliminating manual handoff for supported workflows.
- Built and shipped **chatbot UI/UX** across iOS and Android using Flutter, ensuring **consistent behavior and response rendering** across platforms.

Technologies / Skills Used : Python, LangChain, React.js, Redis, SQL, AWS, RAG, Agentic AI

PROJECTS

Vector AI-Custom Database Engine & RAG Pipeline | Github

Apr 2026 - Present

Lead Developer

- Algorithmic Development:** Developed three distinct Approximate Nearest Neighbor (ANN) algorithms—**HNSW, KD-Trees, and Brute Force**—in **Python**. Optimized search complexity from **O(N) to O(log N)**, achieving sub-millisecond querying for large datasets.
- Database Architecture:** Created a fully custom Vector Database for tracking **768-dimensional embeddings** without using external mathematical libraries. Deployed via a **FastAPI** backend, exposing seven RESTful endpoints for real-time CRUD operations.
- LLM Integration:** Engineered a local Retrieval-Augmented Generation (RAG) system that tokenizes documents into overlapping **250-word chunks**, providing context-aware answers using a locally-hosted **3-billion parameter** language model (**Llama 3.2**).

Technologies / Tools Used : Python, FastAPI, Llama 3.2, React, Tailwind CSS, HTML

DPI Engine - Deep Packet Inspection System | Github

Feb 2026 - Present

Lead Developer

- Multi-threaded Architecture & System Design:** Developed a multi-threaded **Deep Packet Inspection (DPI)** engine in **C++**. Designed a pipeline with Reader → 2 Load Balancers → 4 Fast Path worker threads. Implemented consistent 5-tuple hashing for per-flow affinity and created a custom thread-safe bounded queue (capacity: 10,000 packets) featuring dual condition variable back-pressure and zero-copy packet ownership transfer using **std::move**. Processed TCP, UDP, and DNS traffic across 6 concurrent threads, incorporating per-thread atomic statistics and graceful shutdown sequencing.
- Protocol Parsing and Network Security:** Developed binary-level parsers for 4+ protocols (TLS 1.0–1.3 Client Hello, HTTP/1.x, DNS, QUIC) by traversing raw Ethernet/IP/TCP byte offsets (14 + 20 + 20 bytes) to extract **SNI** hostnames and HTTP Host: headers. Classified traffic into 15+ application categories (e.g., YouTube, Facebook, Google, DNS) using policy-based filtering (block-by-IP, block-by-app, block-by-domain). Generated valid **PCAP** files for inspection in **Wireshark**.

Technologies / Tools Used : C++, Wireshark

WalSmartAI– AgenticAI E-CommercePlatform | Github

Jun 2025 - Sep 2025

Lead Developer

- AI-Driven Search System:** Developed a **Next.js 14** search engine integrated with **Google Genkit** and **Gemini 2.5** via RESTful APIs. Processed natural-language queries across a catalog of over **35 SKUs**, achieving sub-**200ms** latency through optimized data structures and search algorithms, enhancing search relevance by **40%** with schema validation using **Zod**.
- Context-Aware Recommendations:** Designed a server-side LLM retrieval agent that maps user intents to product bundles of **5 to 10** items in a single pass. Leveraged advanced algorithms to reduce time-to-cart by **3 times** and minimize manual filtering for complex queries.
- Sentiment Analytics & Q&A:** Established a real-time sentiment analysis system processing over **1,000 reviews** to generate three key product scores (Quality, Comfort, Fit), visualized with **Recharts** to support data-driven purchasing decisions.

Technologies / Tools Used : Next.js, Google Genkit, Gemini 2.5, Zod, Recharts, React, Tailwind CSS, Express

SKILLS

Additional Skills :	REST APIs, distributed systems
Databases :	MongoDB, Vector Databases (Pinecone), PostgreSQL, MySQL, Redis
Frameworks & Libraries :	HTML5, CSS3, React.js, Node.js, Express.js, Next.js, FastAPI
Programming Languages :	Java, C, JavaScript, Python, C++, Go
AI/ML Skills :	Agentic Workflows, RAG (Retrieval-Augmented Generation), LangChain, LangGraph
Tools & Platforms :	Git, GitHub, Redux, Docker, Jira, Postman, Linux

EDUCATION

New Horizon College Of Engineering | Bangalore

Sep 2023

Bachelor of Engineering

CGPA : 9.19

- Computer Science and Engineering

AWARDS & ACHIEVEMENTS

- Secured Global Rank 549:** in TCS Code Vita Season 12 among 100,000+ participants worldwide, demonstrating strong algorithmic problem-solving skills.
- Completed an intensive Deep Generative AI workshop at IIT Dharwad:** Gaining hands-on experience with modern generative models and deep learning workflows.
- Built an AI-powered KYC verification system:** Using computer vision and NLP; Top 10 Finalist at Hackverse Hackathon among pan-India teams.
- Walmart Sparkathon :** Top 10 Finalist at Walmart's flagship Sparkathon hackathon among pan-India teams; built WalSmart AI — an agentic e-commerce platform with LLM-powered search, product bundling, and sentiment analytics, later developed into a full production system.